# AR-Mini · System Architecture v4

As wired on 31.10.2026 for the water runs: one Olimex ESP32-S3-DevKit-Lipo on the 3S LiPo, the XIAO is the camera node · tracks off for the water · power fused per branch · dashed = idle, designed or on order

## Power




**Turnigy 3S 2200 mAh**
12.60 V full · 4.20 V/cell · C058

12 V



**Inline fuse 30 A**
GT4652 · on the XT60 lead

12 V




**Main switch 50 A**
R13-401A2A · cuts everything

12 V



**6-slot fuse block**
branches below · card C026

**Fused branches (12 V)**

- F1 20 A — thruster (ESC inside the lead)
- F2 15 A — QIQIAZI 5 V / 5 A → lift servos
- F3 — — empty: tracks off for water
- F4 5 A — MZHOU 5 V → DevKit + VBAT sense



**MZHOU buck**
12 → 5 V 3 A · USB-C

F4 5 A
into the DevKit OTG

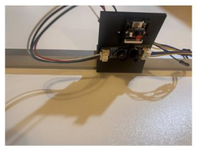


**QIQIAZI buck**
12 → 5 V 5 A · lift

F2 15 A
→ PCA9685 V+ (never 12 V into V+)

On the LiPo since 31.10 (C059); balance charged in a bag on SkyRC iMAX B6 V2. Stop the run at VBAT ≤ 10.5 V.

## Sensing



**Camera head on the arch**
printed mount · 3 cameras · C047



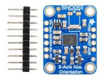
**MLX90642-D45**
45° · wired, idle



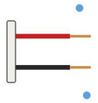
**MLX90640 110°**
wired, idle · C029



**OV3660 2 MP**
XIAO · own Wi-Fi



**BNO055 IMU**
not wired yet



**Leak probe**
GPIO7 · in box

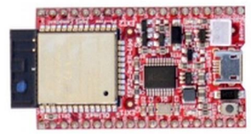


**VBAT divider**
100k / 30k · F4

I²C-1 · I²C-0 · I²C-0 → PCA9685 · VBAT · LEAK

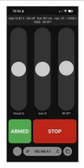
## Control

GPIO4 VBAT
GPIO7 LEAK
GPIO14 KILL rocker
GPIO8 / 9 I²C-1
GPIO48 / 47 I²C-0
USB-C OTG 5 V in
USB-C UART laptop



**Olimex ESP32-S3-DevKit-Lipo**
two breadboards in the food box

GPIO1 / 2 ESC / rudder
GPIO39–42 tracks (off)
Wi-Fi AP "AR-Mini" phone



**Phone remote**
thrust · turn · lift · VBAT on screen



**XIAO ESP32-S3 Sense**
camera node · own AP "AR-Mini-cam"

**Firmware · PlatformIO**
env remote_pool · C059



**74HCT125**
3.3 → 5 V

*Mission state machine (firmware)*

SEARCH → APPROACH → ALIGN → LIFT → RETURN

any state → SAFE on KILL · LEAK · LOW BATTERY · lost heartbeat · 2 s dead-man (thruster 1500 µs, rudder centred, cradle holds)

Stops in series: the KILL rocker puts the firmware in SAFE (1500 µs, rudder centred), S1 cuts the battery, every branch has its own fuse, and a 10 k pull-down holds the ESC signal low through boot. Tested: KILL stops the thruster within 40 ms (C025), and on the battery (C059).

Logs: CSV over USB on the bench (C027 calibration, C050 currents) · 59 loop cards in docs/LOG.md · FAILS_AND_FIXES.md 180 entries

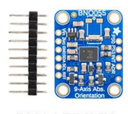
## Next · own PCBs, level shifter, v2 (2027)



**XIAO breakout rev 0**
built · AISLER



**Carrier rev B**
designed · C032



**BNO055 IMU**
heading hold



**WROOM-1 N16R8**
on the carrier



**Raspberry Pi 5**
v2 2027 · ROS 2



**Active cooler**
v2 2027

## Actuation & rescue



**KILL rocker · SPDT**
C → GPIO14 · RUN 3V3 · KILL

EN

F3 empty



**BTS7960 L**
C050 PASS · off for water

EN

F3 empty



**BTS7960 R**
C051 PASS · off for water

M+ M−



**ROB0154 tracks**
built 25.10 · off the boat

→ GPIO14 → SAFE

Tracks: 0.08 A at 20 %, 0.34 A at 80 % on the bench (12 V, free-running) · for the water runs the chassis is off and F3 is empty

F1 20 A

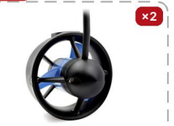


**Cuifati 560 KV · centre**
works · 0.58 A · C025/C059



**Cuifati · right**
faulty · FAILS 173

F1 20 A



×2

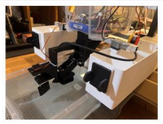
**Blue Robotics T200**
planned replacement




×2

**Basic ESC R3**
1500 µs stop · 3.3 V signal

GPIO1 → 74HCT125 1A → 1Y → ESC signal (5 V, not inverted)



**beam_leg mount**
centre, under beams · C056



**MG90S rudder**
NACA 0015 · 2Y · C057



**Water test · 31.10**
on the battery · C059



**Pesho · thermal dummy**
red PLA · 37 °C · C021

F2 15 A → QIQIAZI 5 V

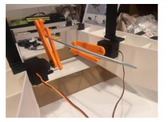



**PCA9685**
0x40 · CH12/13 · V+ 5 V




×2

**TD-8120MG**
lift slider 50–130°



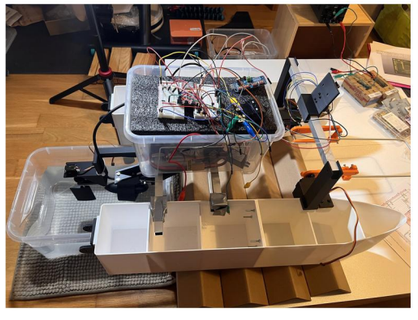
**Parallelogram arms**
Ø5 cradle U · C023



×4

**ST3215 bus servo**
option B · spare

OPEN (FAILS 180): only the higher PCA channel drives a servo; the next run lifts with one servo on CH13.



**The robot as wired · 31.10 (C059)**
hulls, beams, food box, beam_leg, thruster, rudder, cradle

**What is tested (card → PASS):**

- C023 cradle lifts Pesho (mirrored servo pair)
- C025 thruster at 12 V, KILL → 1500 µs in 40 ms
- C027 VBAT reading ±0.02 V from 10 to 12.6 V
- C028 whole robot floats, phone drive over Wi-Fi
- C047–C049 camera head: D45 sees a 39.8 °C head at 1.3 m
- C050 / C051 both tracks run from the BTS7960s
- C055 right thruster faulty, confirmed on a Mega
- C058 LiPo balance charge to 12.60 V
- C059 thruster + rudder in water, on battery, from the phone

**Open:**

- lift on one PCA channel only (FAILS 180)
- lake test and full rescue cycle in water (P22)
- replace the faulty right thruster
- own PCB: carrier rev B (C032)

**Wires:** 12 V (fused branch) · 5 V · I²C-0 (Wire) · I²C-1 (Wire1) · PWM / ESC signal · analog · KILL / EN · motor power · USB · Wi-Fi

Dashed card = idle, designed or on order. Solid wire = wired and tested (card numbers = tests in docs/LOG.md). Common GND everywhere; 12 V reaches the breadboard only on the thin VBAT sense wire (F4, ≈100 µA).

AmphiReach · AR-Mini 1:9 demonstrator · architecture v4 (as wired 31.10.2026; v1 06.10, v2 27.10, v3 28.10 in docs/diagrams/) · photos: BOM listings + build cards · T. Kyoseva, Sofia